FIGHTEX — Economic Simulation Architecture

Engineering Case Study

Public companion to the **fightex-economic-simulation-architecture** repository · English edition · synthetic illustrations only

1. Architecture Overview

FIGHTEX is a Python simulation architecture for a fantasy-sports collectible game economy. It models player behavior, virtual cards, staking, crafting, marketplace trading, tournament payouts and platform finances day by day, and exposes the same engine to analysts (dashboard, CLI) and to AI assistants (MCP). The private implementation, customer parameters and audit results are intentionally not part of this document.

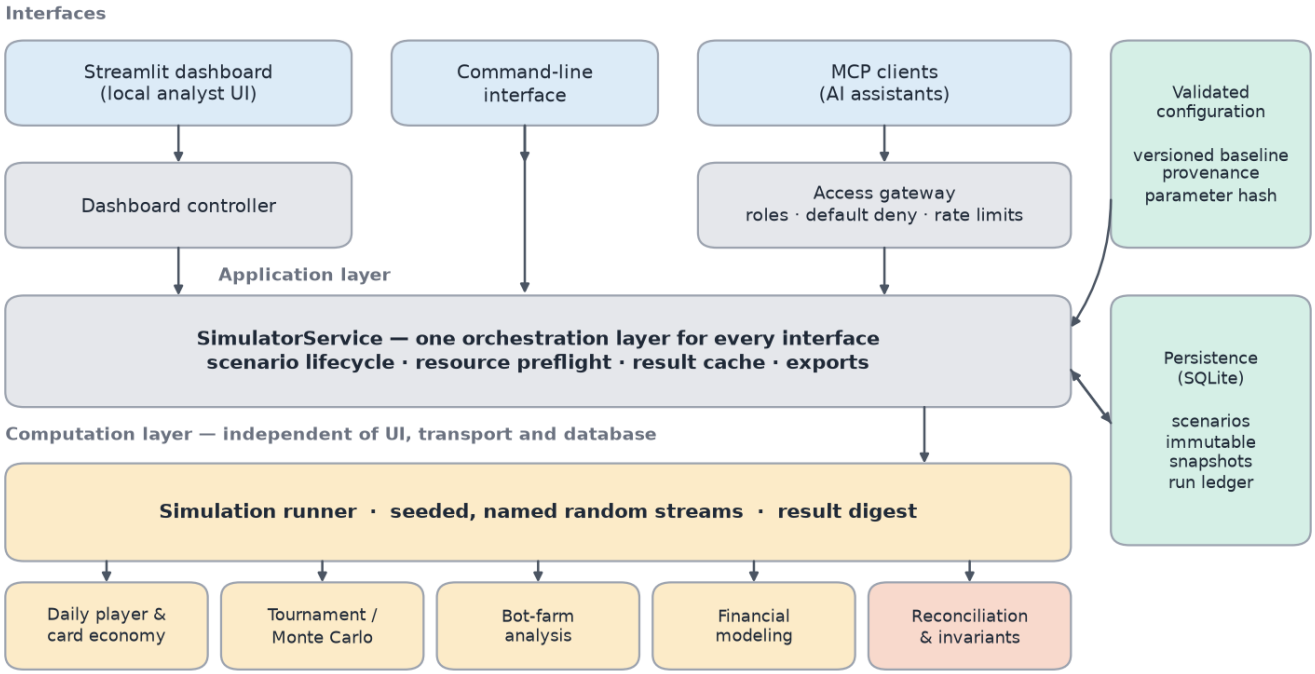


Figure 1. Component diagram (conceptual). Colors group interfaces, orchestration, computation and persistence.

Design decisions that shape the system

Decision	What it means in practice
One engine, many interfaces	Dashboard, CLI and MCP converge on one application service and one runner, so formulas cannot diverge between channels. An AI assistant interprets structured results; the arithmetic stays in Python.
Evidence over narrative	Every number is tied to a versioned configuration, a seed and a result digest. Inputs distinguish customer-defined rules from unverified behavioral assumptions.
Default-deny access	Roles are assigned by the server, never by tool arguments. Tools and data are exposed through an explicit allow-list; an operational read-only mode can disable all calculations.
Resource-aware execution	Requests are validated and cost-estimated before they run; oversized requests are rejected instead of being started.

Technology: Python 3.12+, NumPy, SciPy, Pandas, Pydantic, SQLite, Streamlit, pytest and Hypothesis, the Python MCP SDK; Docker Compose and Nginx for server deployment.

In this document

Page	Topic	Visualization
2	Multi-account simulation	Shared-state diagram; order flow; milestones
3	Marketplace liquidity	Supply / demand / executed trades
4	Bot-farm economics	Three valuation lenses; liquidation curves
5	Financial modeling	Revenue / expenses / cash flow
6	Sensitivity analysis	Tornado chart; seed-spread plot
7	Reproducibility	Seed → simulation → digest → audit

2. Multi-account Simulation

Coordinated multi-account behavior is only meaningful inside the economy it exploits. The bot-farm analysis therefore does not use a stand-alone ROI spreadsheet that assumes unlimited buyers: multi-account actors are a slice of the **same simulated population**, running through the **same daily state** as organic players — the same card supply and expiry, order book, staking ledger, tournament pools, fees and withdrawal rules.

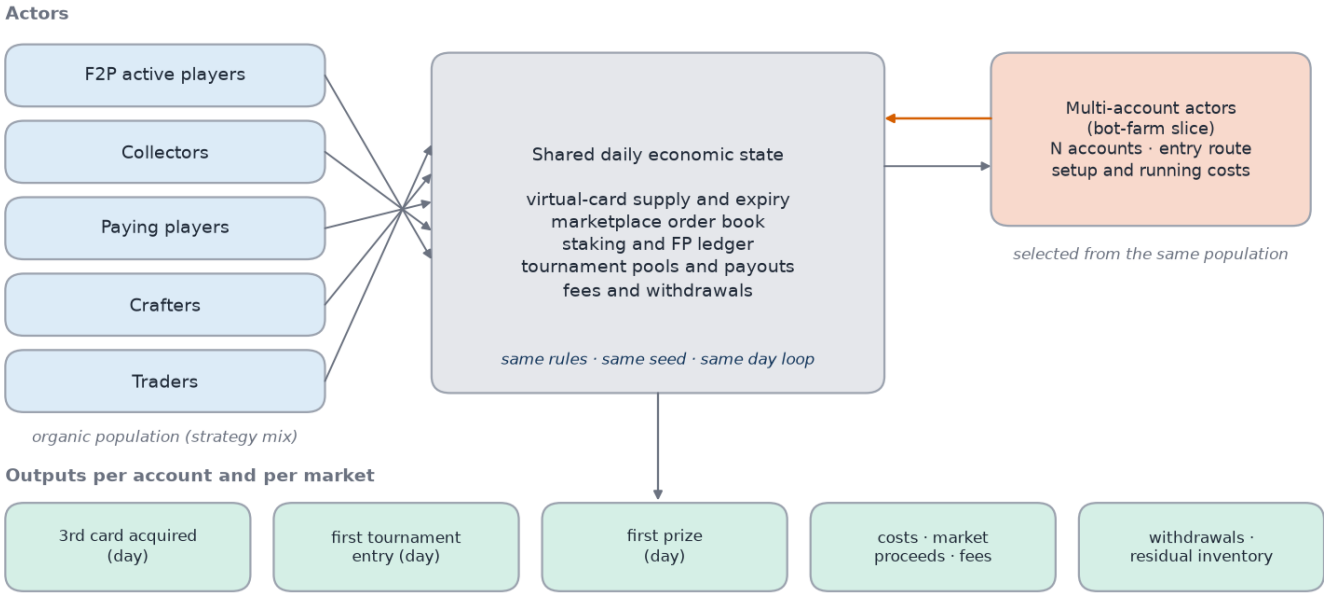
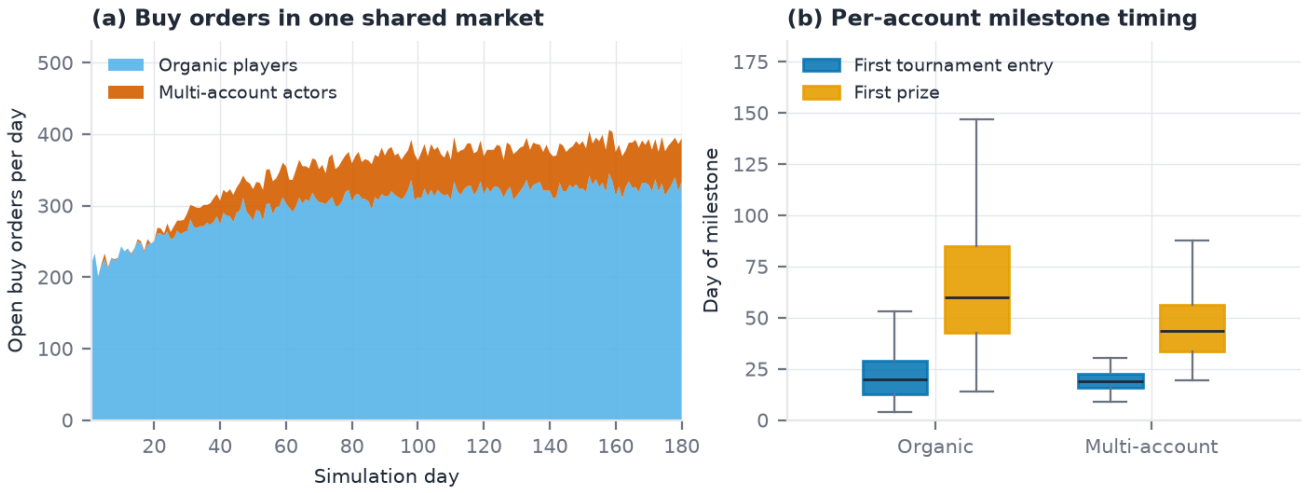


Figure 2. Organic and multi-account actors share one economic state (conceptual diagram).



Illustrative synthetic data — not actual FIGHTEX audit results.

Figure 3. Illustrative output of a shared-economy run: (a) both groups place orders in one market; (b) per-account milestone timing. Illustrative synthetic data — not actual FIGHTEX audit results.

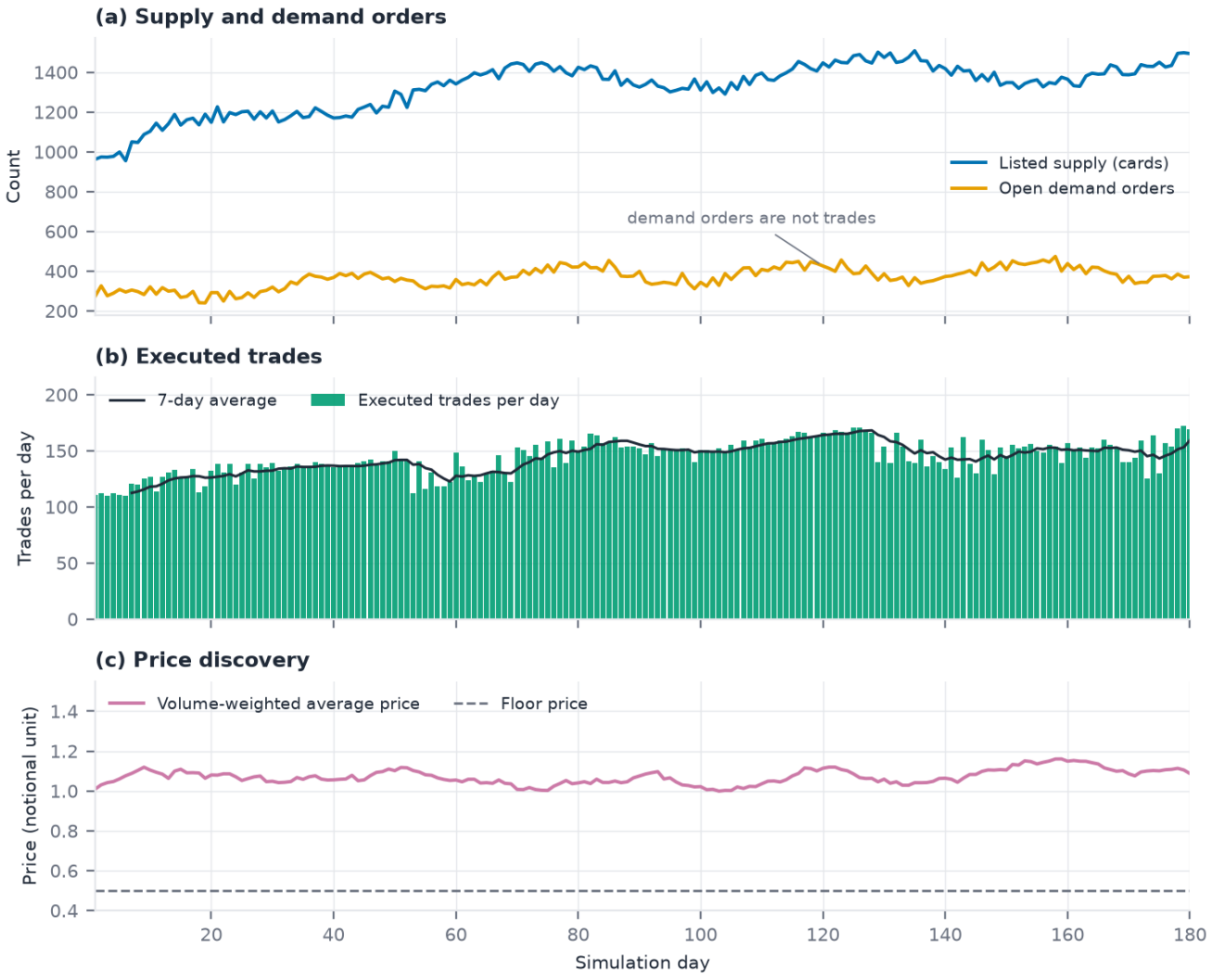
What the analysis can answer

- **Per-account milestones:** day of the third card, first tournament entry and first prize, plus operating and setup expenses.
- **Market feedback:** multi-account demand competes with organic demand, so purchases, fees and sell-offs change the prices the same accounts later face.
- **Entry-route comparison:** alternative ways to acquire cards can be compared without publishing proprietary behavioral coefficients.
- **Distribution, not just a mean:** per-account outcome distributions show how much of a result depends on a few lucky accounts.

3. Marketplace Liquidity

A card is only worth what the market can absorb. The marketplace analysis separates what is *offered*, what is *wanted* and what actually *trades*, and it does not assume that market-wide demand is exclusively available to one seller.

Quantity	Definition used in the analysis
Listed supply	Cards currently offered for sale, by rarity.
Demand orders	Open buy intent at acceptable prices. It is an upper bound on trades, not a trade count.
Executed trades	Matches between a listing and a demand order within a day; the only flow that creates fees and proceeds.
Volume-weighted price	Average executed price weighted by volume; compared with the floor price to see price discovery and depth.



Illustrative synthetic data — not actual FIGHTEX audit results.

Figure 4. Supply, demand orders, executed trades and price for one rarity over 180 simulated days. Illustrative synthetic data — not actual FIGHTEX audit results.

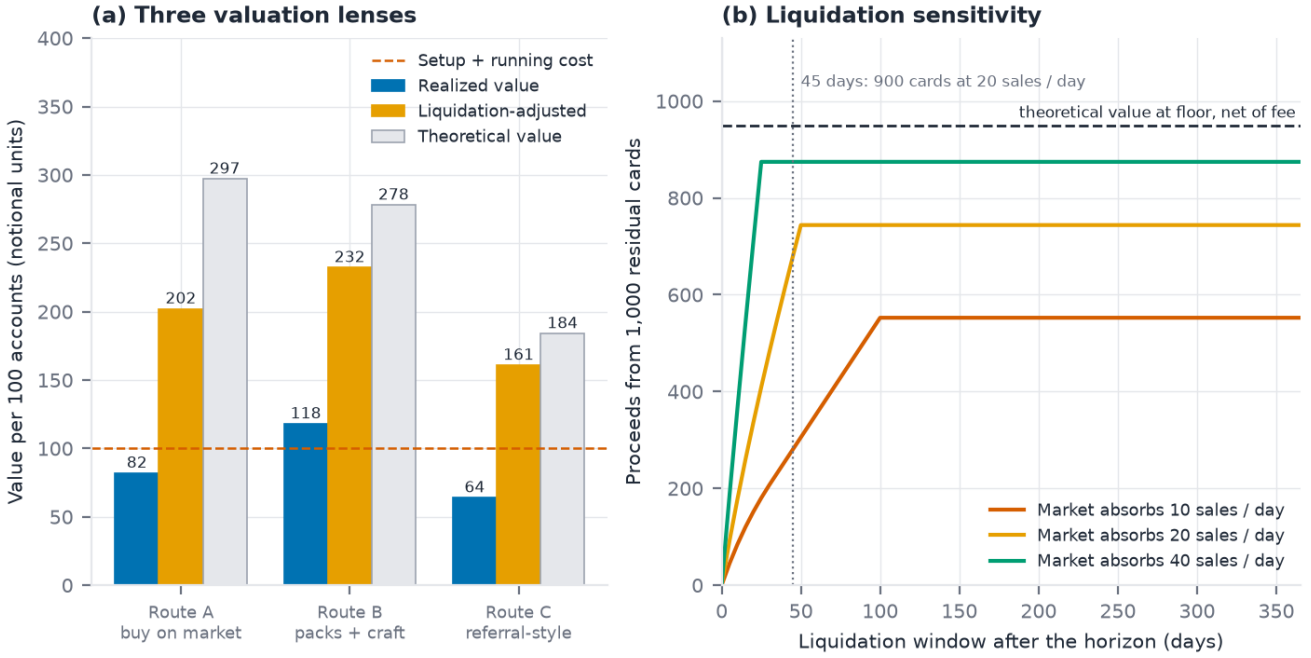
Reading the chart

- In this illustration listed supply is about three to four times larger than open demand orders, so only a small share of listed cards can sell on any given day.
- Executed trades are bounded by both sides of the book, so a doubling of demand orders does not double proceeds.
- Executed volume per day is the quantity that later limits how fast a residual inventory can be sold (next section).

4. Bot-farm Economics

A multi-account strategy can look profitable on paper while most of its value is still locked in cards that have to be sold into a finite market. The analysis therefore reports three lenses side by side instead of one ROI figure.

Lens	What it represents
Realized value	Cash actually withdrawn, after acquisition, setup and operating costs.
Theoretical residual value	Remaining live cards marked at the floor price, net of the marketplace fee.
Liquidation-adjusted outcome	Result after trying to sell the residual inventory during a finite liquidation window, at the depth the market can absorb.



Illustrative synthetic data — not actual FIGHTEX audit results.

Figure 5. (a) The three lenses for three illustrative entry routes; (b) liquidation-adjusted proceeds of 1,000 residual cards vs. the length of the liquidation window for three market depths. Illustrative synthetic data — not actual FIGHTEX audit results.

Liquidation sensitivity

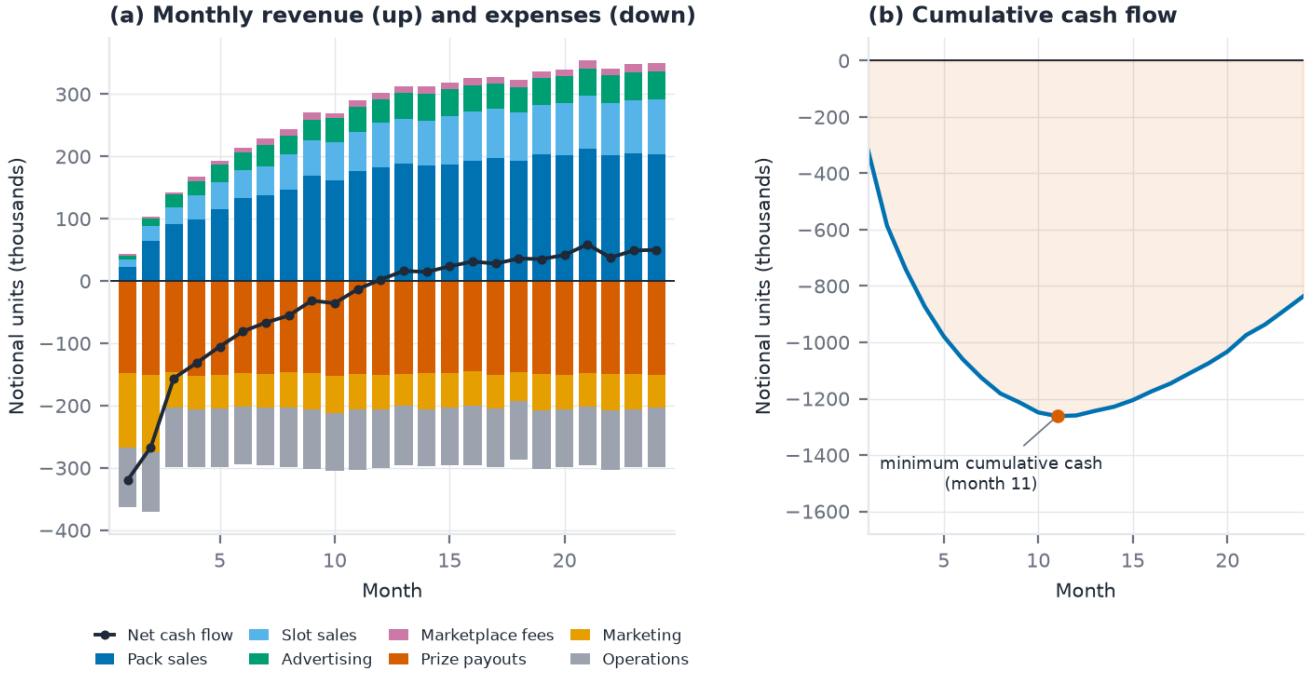
Marking 1,000 cards at 1.00 gives a notional 1,000 (950 after a 5% fee). If the market absorbs 20 sales per day and the seller captures all of them, selling 900 cards takes at least 45 days — and only if demand stays stable. In this synthetic example a 45-day window in a market that absorbs 20 sales per day yields about 680 of the 950 theoretical value: about 100 cards are still unsold and the seller has pushed its own price down. The engine continues the same economy after the horizon (a liquidation window), so competition, fees and expiry keep acting on the residual inventory.

Liquidation assumption explored	Effect on the outcome
Length of the liquidation window	Too short a window leaves cards unsold; beyond the sell-down time more time adds nothing.
Market depth (sales per day the market absorbs)	Sets both the sell-down time and how far the seller pushes the price.
Marketplace fee	A fixed haircut on every executed sale.
Price impact of the seller's own supply	Selling into a thin book lowers the price of the next sale.
Card lifetime during the window	Cards that expire before they are sold are worth nothing.
Competing sellers	Other listings share the same demand, so the farm does not capture all executed trades.

5. Financial Modeling

The financial layer turns the simulated flows into three views that must reconcile with each other, rather than one profit number.

View	Question it answers
Accrual result	What did the platform earn and spend in the period, regardless of when cash moved?
Cash flow	When does money actually move? Drives the cumulative cash curve and the minimum cash position.
Obligations and reserves	What is owed to players (balances, prizes) and what reserve is required to stay solvent?



Illustrative synthetic data — not actual FIGHTEX audit results.

Figure 6. Illustrative 24-month cash-flow model: (a) revenue and expense categories with net cash flow; (b) cumulative cash and its minimum. Illustrative synthetic data — not actual FIGHTEX audit results.

Controls built into the financial engine

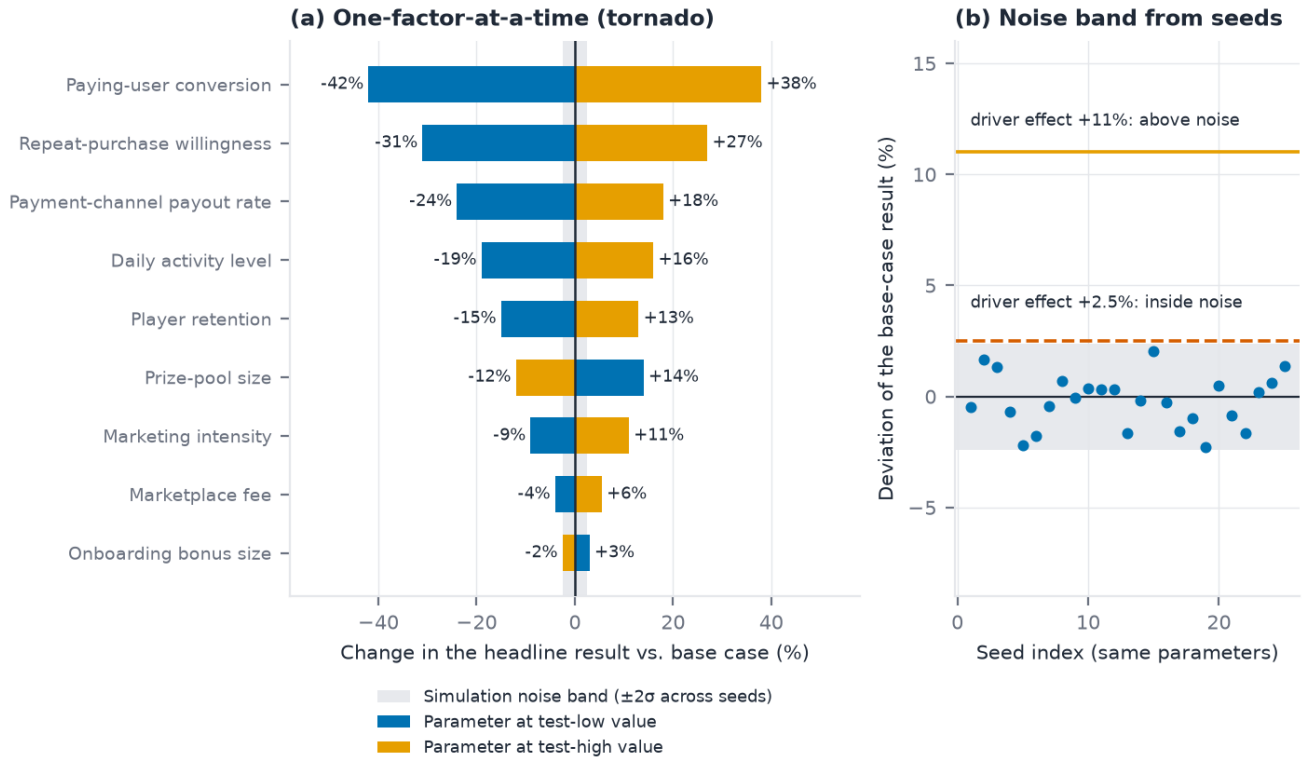
- **Reconciliation identity:** cash + receivables − liabilities − unallocated amounts must equal the cumulative result; mismatches fail the run's checks.
- **Horizon coverage invariant:** aggregation must account for exactly the simulated number of days. After an independent review, regression tests cover horizons that are not multiples of 7 (for example 8, 9, 15, 729 and 730 days), so a trailing partial week cannot be dropped silently.
- **Capital need is a minimum, not a total:** the largest cumulative cash deficit and required reserves are reported separately from profit (here the synthetic minimum is 1,262 thousand notional units below zero, in month 11).
- **Scenario honesty:** payment-channel fees, payout rates and marketing are scenario parameters; results are conditional on them.

Examples of reconciliation checks

Check	What it catches
Purchases counted once	A purchase paid in one currency or channel must not be counted twice in revenue.
Prize payouts equal engine payouts	Money credited to players in the tournament module equals the expense and the liability in the financial layer.
Balance-sheet style identity	Cash, receivables, obligations and unallocated amounts must add up to the cumulative result.
Every day of the horizon is covered	No simulated day may be missing from the weekly or monthly financial tables.

6. Sensitivity Analysis

Many inputs of a game economy are assumptions rather than measurements. Sensitivity analysis shows which assumptions the conclusion actually depends on, so that verification effort goes where it matters.



Illustrative synthetic data — not actual FIGHTEX audit results.

Figure 7. (a) Effect of moving one driver at a time to a low and a high test value (generic driver names); (b) the spread of an unchanged base case across seeds defines the noise band. Illustrative synthetic data — not actual FIGHTEX audit results.

Method element	How it is used
One factor at a time	Each driver is moved alone, with the same seed and engine, so the effect is attributable.
Test ranges, not business values	Low and high values (for example ×0.5 and ×1.5) are probes; they are not claims about acceptable or likely values.
Noise threshold	Effects inside the simulation-noise band (about 2 standard deviations across seeds) are not reported as influential.
Conclusion flips	The analysis flags drivers that change the sign of a result or the severity of a rule-based finding, not only its size.
Criticality ≠ risk	A parameter's influence is kept separate from the criticality of the business risk it touches.

How the result is used

- **Rank, then verify:** the longest bars are the assumptions to confirm with real data first; short bars can stay assumptions without changing the conclusion.
- **Compare with noise:** an effect is only meaningful when it exceeds the spread of the unchanged base case across seeds (panel b).
- **Stay conditional:** a driver that dominates under one set of assumptions need not dominate under another, so the analysis is repeated when the baseline changes.

7. Reproducibility

Seed → simulation → digest → audit: every result can be traced to the exact inputs and software context that produced it, and a repeat run must produce the same digest.

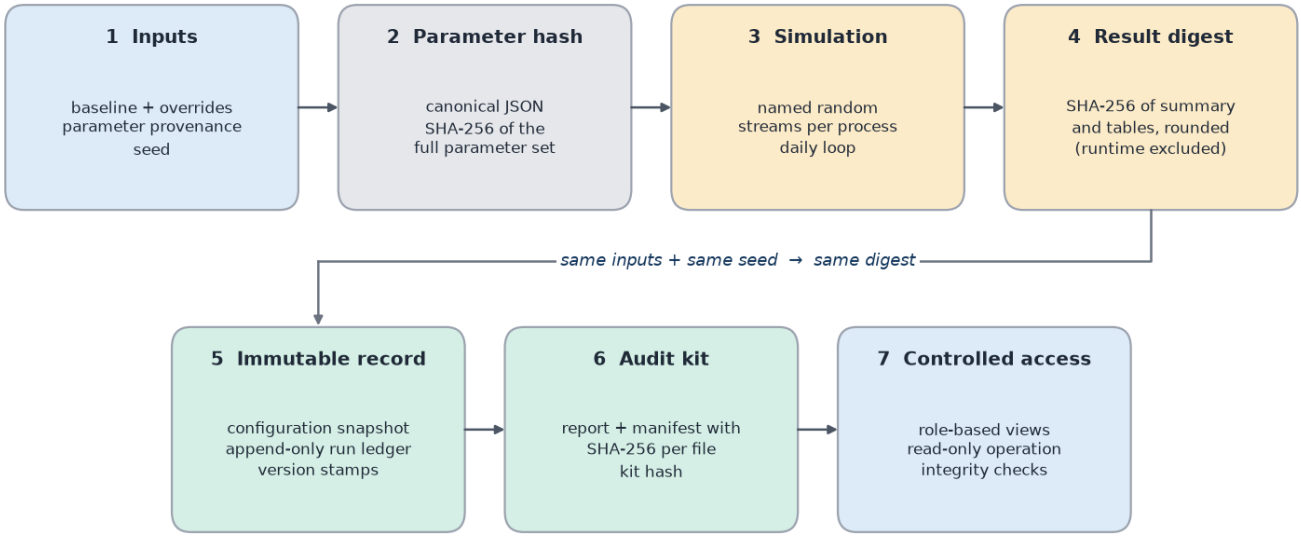


Figure 8. Reproducibility pipeline (conceptual).

Recorded for every run	Purpose
Engine and configuration version, parameter hash, seed	Identify exactly what was simulated.
Input-data provenance (synthetic or structured sports data)	Separate modeled inputs from real data.
Result digest (summary + tables, runtime excluded)	Detect any change in the numbers, however small.
Immutable configuration snapshot and append-only run ledger	Keep the basis of a conclusion after scenarios change.
File-level SHA-256 manifest of the audit kit	Prove that the delivered documents match the verified calculations.

Determinism demonstration

The table below is produced by this script with a deliberately tiny seeded model (60 days, three named random streams). It is **not** the FIGHTEX engine; it only shows the mechanics: the same seed and parameters give the same digest, while a different seed or a single changed parameter gives a different one.

Run	Seed	Parameter hash	Result digest	Same as Run A?
Run A	42	`97b77e371695886b`	`340799de00c77160`	yes
Run B (repeat of A)	42	`97b77e371695886b`	`340799de00c77160`	yes
Run C (different seed)	43	`97b77e371695886b`	`9585099410f5d648`	no
Run D (one parameter changed)	42	`e200dd55f2fa7aa8`	`d041cf5715d1694d`	no

Illustrative synthetic data — not actual FIGHTEX audit results. Digests are computed from the toy model at build time.

Scope and limitations

Simulation outputs are conditional scenarios, not financial forecasts. Independent review findings are tracked and re-validated; corrected calculations ship as a new, separately identified audit kit rather than overwriting an accepted one. This document is not a security certification.